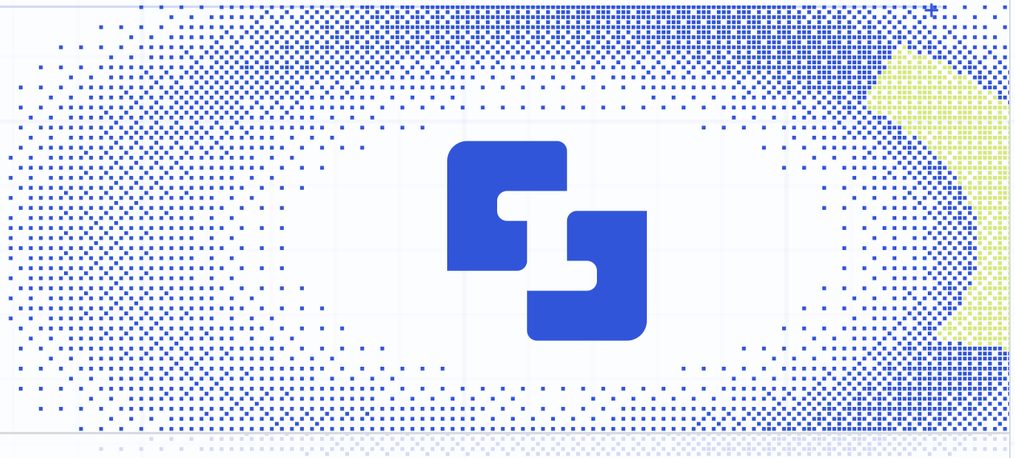
Maker Day 3

BenchMate

Scalable, safe training,
from expert to novice.



From MakerLAB To Manufacturers

Our customer: organizations that train people for hands-on work.

01 OUR WEDGE · NOW

Cornell Tech MakerLAB

Novice makers

PILOT UNDERWAY

→

02 NEXT

Technical Colleges

Trade students

PLANNED · CONTACT COLLEEN DEMPSEY

→

03 THEN

Manufacturers

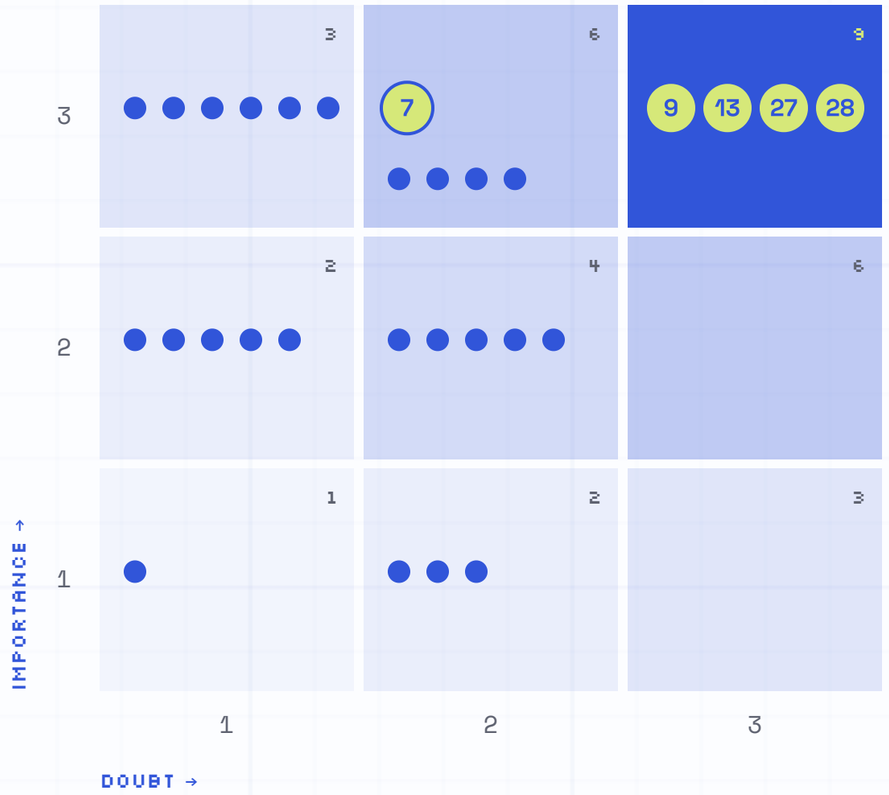
Employee training teams

PLANNED · FIRST INTERVIEW

WHY THIS ORDER Prove It, Earn Trust, Then Sell.

Our Tallest Tales

29 hypotheses scored. Risk factor = doubt × importance.



TOP FIVE BY RISK FACTOR

9 — 3 × 3

#9 Learners Finish Tasks ≥30% Faster

Core value: faster learning drives adoption.

EXPERIMENT 3

9 — 3 × 3

#13 Organizations Share Proprietary SOPs

Enterprise adoption needs proprietary sources.

9 — 3 × 3

#27 AI Responds Fast Enough For Real-Time Help

Slow answers break hands-on learning.

9 — 3 × 3

#28 Learners Trust AI Guidance

Without trust, learners will not follow.

EXPERIMENT 2

6 — 2 × 3

#7 AI Parses Raw Video Into Guidance

Foundation for scalable tutorial creation.

EXPERIMENT 1

Three Experiments To De-Risk BenchMate

Thresholds set before each run.

01 / HYPOTHESIS #7

Video To Instructions

EXPERIMENT	Prototype · Technical Benchmark
BEHAVIOR	Steps match expert labels

PASS IF

≥80% accuracy, 0 critical omissions

02 / HYPOTHESIS #28

Learner Trust

EXPERIMENT	Prototype · Wizard Of Oz
BEHAVIOR	Chooses to follow the AI

PASS IF

≥3 of 4 follow the guidance

03 / HYPOTHESIS #9

Time Saved

EXPERIMENT	Pilot · A/B Test
BEHAVIOR	Completes the task, timed

PASS IF

≥30% time saved, ≥80% pass

Path: Sequence & Constraints

Ordered by risk reduction per team-person hour, within constraints.

01

Accuracy & Safety

FINISH

Nov 17

19 h

Needs 4 uncut videos and expert labels

→

02

Response Time

FINISH

Nov 18

4 h

Needs a stable API

→

03

Learner Trust

FINISH

Nov 18

~6 h

Needs checked guidance from step 01

→

04

MakerLAB Pilot

FINISH

Nov 19

12–16 h

Needs lab approval and filming consent

FALLBACK Steps 1 To 3 Proceed Even If MakerLAB Access Is Delayed.

Experiment 1: Video To Instructions

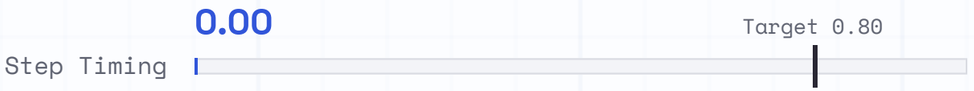
UNDERWAY

HYPOTHESIS #7	EXPERIMENT	BEHAVIOR	PASS IF
AI turns raw video into instructions	Prototype · Technical Benchmark	Extracted steps match expert labels	≥80% accuracy, 0 critical omissions

WHERE IT BREAKS · UNCUT VIDEO

22 steps generated vs **7** reference





WHAT HOLDS · BOTH VIDEOS

	Human-edited · hand-washing	Uncut · X1C printer task
Facts Found*	6/6	7/7
Gaps Flagged*	2/2	2/2
SOP Checks Found*	1/1	2/2

The human-edited video recovers 4 of 6 steps.

EARLY READ

Facts are captured. Steps on uncut video are split too finely: not yet a pass.

*Keyword checks, not expert judgments; hand poses not yet checked.

Experiment 2: Safety & AI Trust

UNDERWAY

HYPOTHESIS #28	EXPERIMENT	BEHAVIOR	PASS IF
Learners trust AI guidance	Prototype · Wizard Of Oz	Chooses to follow the guidance	≥3 of 4 follow verified guidance

SAFETY · 2 RUNS

100%

Of Original Warnings Kept

CAUTIONS IN THE FINAL TUTORIAL

X1C Wiper Mod ■■■■ □□ ■■■■ 10

PCB Soldering ■■■ ■■■■■■ 9

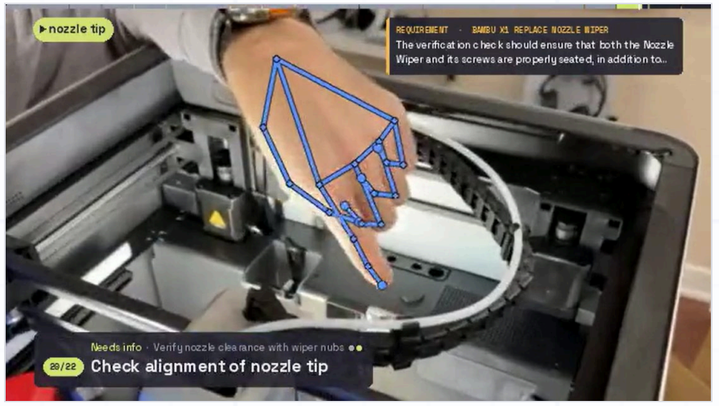
■ From the video □ Expert answers ■ Added by BenchMate

SENT TO THE EXPERT, NOT GUESSED

6 SOP conflicts flagged, medium to high

3 Key steps flagged for expert feedback

5 of 6 SOP steps covered fully, the sixth partly ■■■■■◧



> "I know that this is solving the task I'm supposed to do, not something else AI hallucinated."
>
> BenchMate Learner · Soldering

EARLY READ

Keeps every warning, adds more, and asks the expert when unsure.

One added soldering caution was wrong ("pins" should be "iron"), so expert review stays. The 8-novice trust test finishes Nov 18.

Experiment 3: Expert Time

COMPLETED | FURTHER PARTICIPANTS REQUIRED

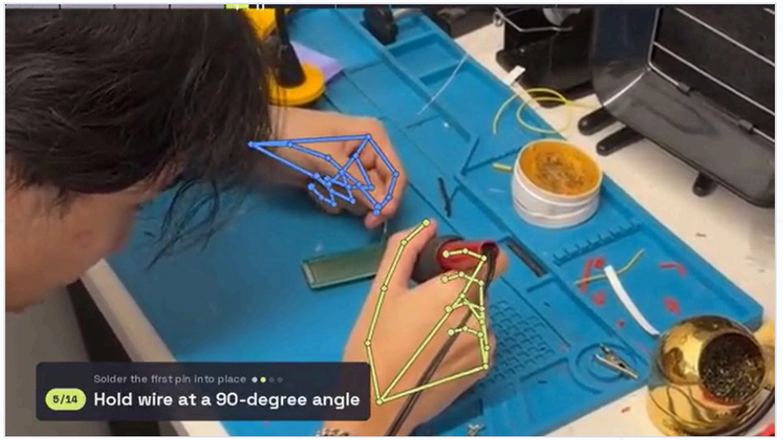
HYPOTHESIS #9	EXPERIMENT	BEHAVIOR	PASS IF
BenchMate cuts training time by ≥30%	Pilot · A/B Test	Completes the task correctly, timed	≥30% saved, ≥80% pass, 0 omissions

EXPERT EDITING TIME

Manual Edit* — 40–60 min

With BenchMate — ~28 min (2 min × 14 steps)

Plus 4:33 to record: about 33 min of expert time in total. Another ~20 min of processing runs on its own.



Edited output: hand poses, step label, crop to the work.

BENCHMATE AUTOMATES

- Timestamped Steps
- Pause Removal
- Crop To Hands
- Summary Cards
- Step Labels
- Safety Cautions
- SOP References
- Text Guide

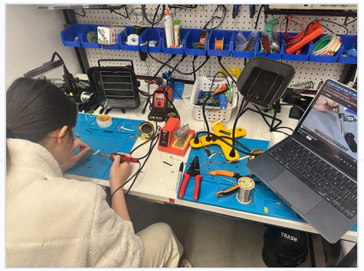
*A low estimate: it grows with unfamiliar editing software and look-ups in SOPs and safety manuals.

Experiment 3: Learner Time

COMPLETED | FURTHER PARTICIPANTS REQUIRED

HYPOTHESIS #9	EXPERIMENT	BEHAVIOR	PASS IF
BenchMate cuts training time by ≥30%	Pilot · A/B Test	Completes the task correctly, timed	≥30% saved, ≥80% pass, 0 omissions

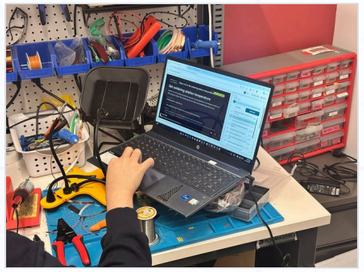
TIME TO LEARN THE TASK



A / YOUTUBE ONLY

19:00

“Some details are in the end of the video, so I need to watch the whole video many times.”



B / BENCHMATE

5:14

“I liked how I could repeat specific steps with the timestamps.”

EARLY READ

72%

Less Learning Time Than YouTube Alone

THRESHOLD: ≥30%

One first-time learner per condition: directional, not yet a pass. YouTube is a first benchmark; the training buyers use today comes next.

Experiment 3: Cost & Quality

COMPLETED | FURTHER PARTICIPANTS REQUIRED

HYPOTHESIS #9	EXPERIMENT	BEHAVIOR	PASS IF
BenchMate cuts training time by ≥30%	Pilot · A/B Test	Completes the task correctly, timed	≥30% saved, ≥80% pass, 0 omissions

EXPERT TIME TO TRAIN 20 LEARNERS

Expert Guidance* — 1 h 31 min

BenchMate — ~33 min one time

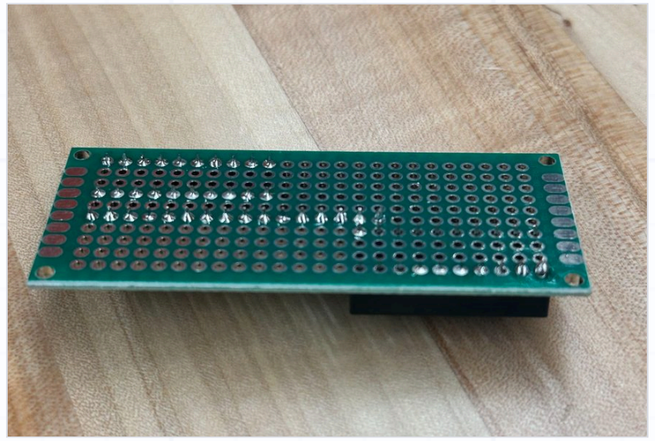
YouTube Only — 0 min but 19:00 per learner, against 5:14

58 min

Expert Time Saved

About 64% less than expert guidance. Threshold: ≥30%.

TASK QUALITY



No discernible change in soldering quality.

*Generous assumption: a learner guided by an expert moves at the expert-only pace of 4:33 each.

What Changed & What We Build Next

What the results so far say about each hypothesis.

01 / EXTRACTION

22 vs 7

steps on the uncut video

Facts Hold, Steps Over-Split

BUILD NEXT

Fix how steps are split; re-test on 4 uncut videos.

02 / TRUST

100%

of original warnings kept

Safety Guidance Survives Editing

BUILD NEXT

Run the 8-novice trust test; keep expert review.

03 / VALUE

72%

less learner time than YouTube alone

Time Savings Look Large

BUILD NEXT

MakerLAB pilot; then test against buyers' own training.

ON THE CANVAS **Technical Colleges Follow MakerLAB; The Buyer Is Manufacturers' Training Teams.**

Small samples so far: directional evidence, not yet a pass.

Thank You

BenchMate

Questions?